

Report

Rationale for Visualisation Technique

Visualisation - 3D map-based bar chart. The 3D map-based bar chart is a visualisation that places the cities of the UK geographically and reveals the geospatial distribution of the information by plotting cities with vertical bars on a map of the UK. This visualisation method was used because the data set contains both spatial and numerical information.

A 3D bar encoding was used, which represented the population value in terms of height. This allowed users to compare relative values between cities quickly. This encoding gave the relative importance of values with respect to each other, I E., the larger the population, the greater the value, which was immediately apparent. To add another layer of comparison, colour encoding (blue to red) was also used.

One of the problems with the data set has been the large variation in population values, particularly with the likes of London, skewing the scale. This was accounted for with the use of a log scale, a process which is common in data visualisation to make smaller cities more visible.

The map-based layout keeps the visualisation geographically accurate and lets the user understand population distributions by location, for example, to understand that high population density appears in clusters in certain regions.

In terms of usability, a few features were added to improve it. Pan and zoom buttons allow the user to virtually fly around the scene and zoom in. There is a button by the years in which, when you click it will allow you to zoom in between the 3 data sets (1991, 2001, 2011) and compare them to each other. The filter button was added because it is based on the population value; therefore, if you want to show only the cities that have a certain population, then you can use that more effectively (not displaying all the cities).

A hover interaction was used so that more detail (name, population, coordinates) could be provided without overloading the visual layout. This technique fits with the “overview first, details on demand” principle.

A colour legend has been added so the reader can interpret the scale of colours used.

The overall approach does successfully integrate the visualisation of location with the quantification of population figures as well as afford interaction, making this a plausible technique for accessing UK population data. Its design adheres to good visualisation practices.

Screenshots

Figure 1: Main 3D Map View

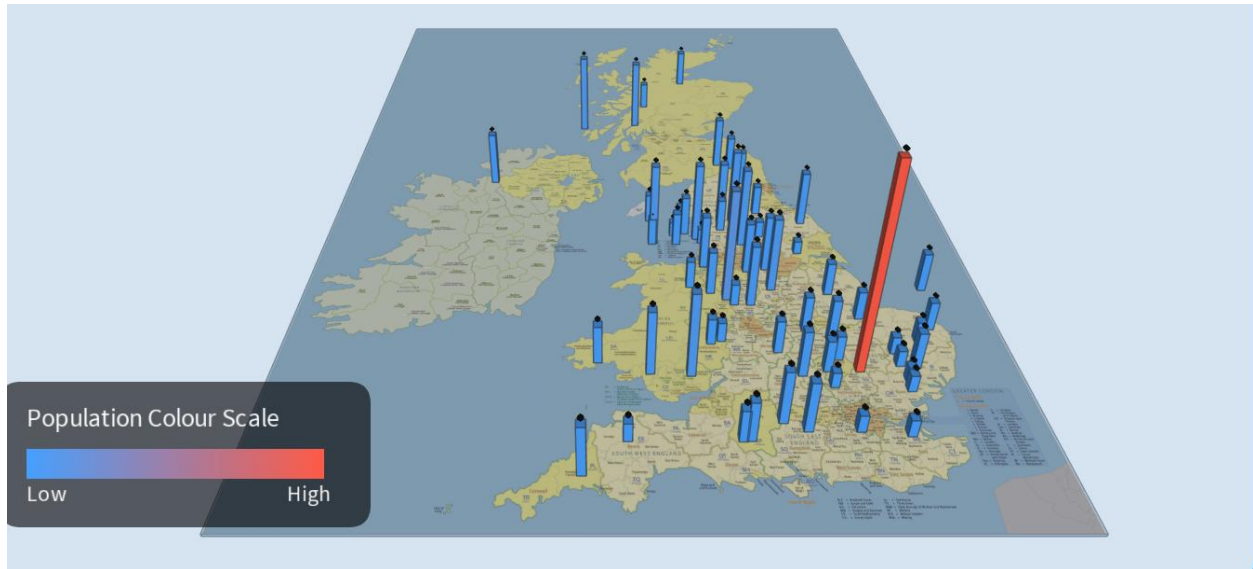


Figure 2: Zoomed and Pan View with hover interaction

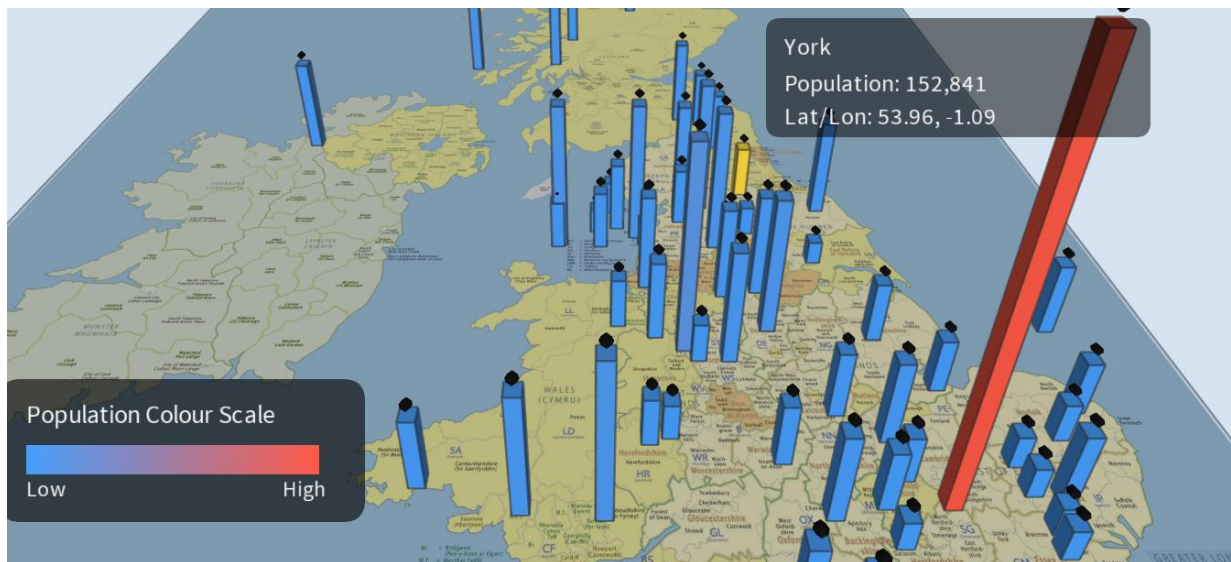


Figure 3: 1991 Population

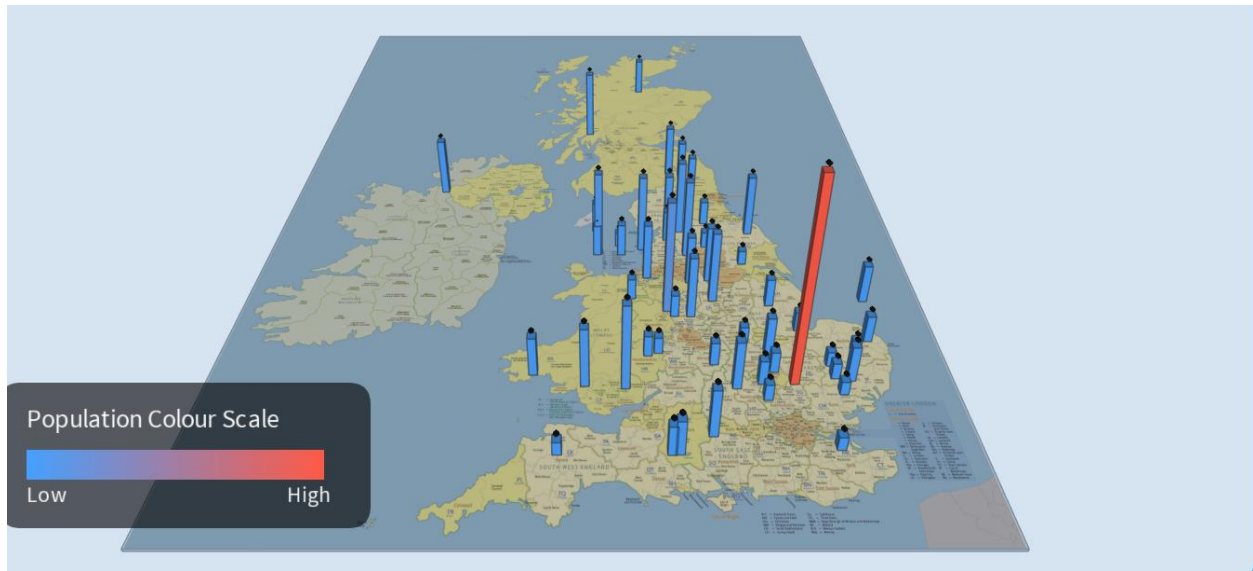


Figure 4: 2011 Population

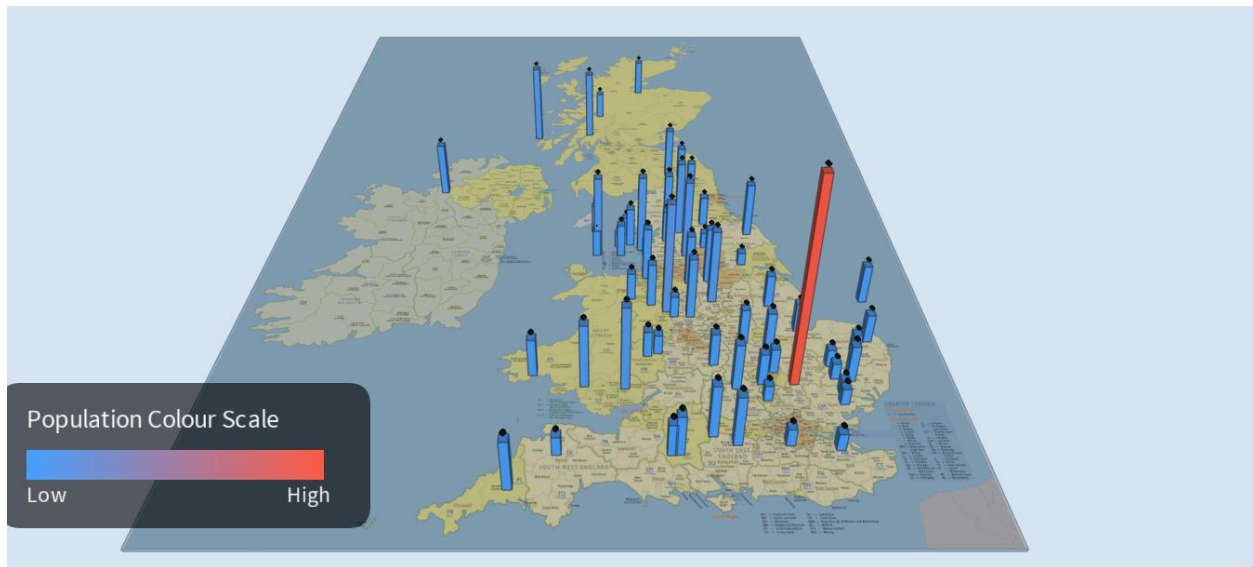


Figure 5: Filtered population view where only cities exceeding the threshold are displayed

